1) **(20 points) Find the radius of convergence and interval of convergence of the following power series.**

(a) $\sum_{n=1}^{\infty} \frac{(-1)^n (x-3)^n}{n^{2/3}\, 4^n}$

Root Test (could also use Ratio Test):

$$\sqrt[n]{|a_n|} = \frac{|x-3|}{n^{2/3n} \cdot 4} \to \frac{|x-3|}{4}$$

converges if $\frac{|x-3|}{4} < 1$

$|x-3| < 4$, so $\boxed{R=4}$

$-4 < x-3 < 4$

$-1 < x < 7$

$x=-1$: $\sum \frac{(-1)^n(-4)^n}{n^{2/3}4^n} = \sum \frac{1}{n^{2/3}}$ diverges

$x=7$: $\sum \frac{(-1)^n 4^n}{n^{2/3}4^n} = \sum \frac{(-1)^n}{n^{2/3}}$ converges

$\boxed{(-1,7]}$

(b) $\sum_{n=0}^{\infty} \frac{(2n)!\,(x-2)^n}{n!}$

Ratio Test:

$$\left|\frac{a_{n+1}}{a_n}\right| = \left| \frac{(2n+2)!\,(x-2)^{n+1}}{(n+1)!} \cdot \frac{n!}{(2n)!\,(x-2)^n} \right|$$

$$= \frac{(2n+2)(2n+1)}{n+1} |x-2|$$

$$\to \begin{cases} \infty & \text{if } x \neq 2 \\ 0 & \text{if } x = 2 \end{cases}$$

converges only when $x=2$

$$\boxed{\begin{matrix} R=0 \\ \{2\} \end{matrix}}$$

2) (20 points) (a) Find a power series representation for the function

$$f(x) = \frac{1}{9-x^2} = \frac{1}{9} \cdot \frac{1}{1-\frac{x^2}{9}}$$

What is the radius of convergence of this series?

$$\frac{1}{9-x^2} = \frac{1}{9}\sum_{n=0}^{\infty}\left(\frac{x^2}{9}\right)^n$$

$$= \boxed{\sum_{n=0}^{\infty} \frac{x^{2n}}{9^{n+1}}}, \quad \left|\frac{x^2}{9}\right| < 1 \Rightarrow |x^2| < 9 \Rightarrow |x| < 3$$

$$\boxed{R = 3}$$

(b) Use the series you found in part (a) to find a power series representation for the function

$$g(x) = \frac{x^3}{(9-x^2)^2}$$

What is the radius of convergence of this series?

$$\frac{1}{9-x^2} = \sum_{n=0}^{\infty} \frac{x^{2n}}{9^{n+1}}, \quad R = 3. \quad \text{Differentiate:}$$

$$-(9-x^2)^{-2}(-2x) = \sum_{n=1}^{\infty} \frac{2nx^{2n-1}}{9^{n+1}}, \quad R = 3$$

$$\frac{2x}{(9-x^2)^2} = \sum_{n=1}^{\infty} \frac{2nx^{2n-1}}{9^{n+1}}. \quad \text{Divide by 2:}$$

$$\frac{x}{(9-x^2)^2} = \sum_{n=1}^{\infty} \frac{nx^{2n-1}}{9^{n+1}} \quad \text{Multiply by } x^2:$$

$$\frac{x^3}{(9-x^2)^2} = \boxed{\sum_{n=1}^{\infty} \frac{nx^{2n+1}}{9^{n+1}}}, \quad \boxed{R = 3}$$

3) (20 points) (a) Use the definition of Taylor series to find the Taylor series of $f(x) = \frac{1}{2x+7}$ centered at $a = 1$. (You do not need to find the radius of convergence.)

$f(1) = \frac{1}{9}$

$f'(x) = -(2x+7)^{-2}(2) \Rightarrow f'(1) = -\frac{2}{9^2}$

$f''(x) = 2(2x+7)^{-3}(2^2) \Rightarrow f''(1) = \frac{2 \cdot 2^2}{9^3}$

$f'''(x) = -6(2x+7)^{-4}(2^3) \Rightarrow f'''(1) = \frac{-6 \cdot 2^3}{9^4}$

$\vdots$

$f^{(n)}(1) = \frac{(-1)^n n!\, 2^n}{9^{n+1}}, \quad n \geq 0$

$\frac{f^{(n)}(1)}{n!} = \frac{(-1)^n 2^n}{9^{n+1}}$

$$\frac{1}{2x+7} = \boxed{\sum_{n=0}^{\infty} \frac{(-1)^n 2^n}{9^{n+1}} (x-1)^n}$$

(b) Evaluate the following integral as an infinite series. (You do not need to find the radius of convergence.)

$$\int x^2 \sin(3x^2)\, dx$$

$$= \int x^2 \left(\sum_{n=0}^{\infty} \frac{(-1)^n (3x^2)^{2n+1}}{(2n+1)!} \right) dx$$

$$= \int x^2 \left(\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n+1} x^{4n+2}}{(2n+1)!} \right) dx$$

$$= \int \left(\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n+1} x^{4n+4}}{(2n+1)!} \right) dx$$

$$= \boxed{C + \sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n+1} x^{4n+5}}{(2n+1)!(4n+5)}}$$

4) **(10 points) Let $f(x) = (3x-1)^{5/3}$**

(a) Approximate f by a Taylor polynomial with degree 2 centered at $a = 3$.

(b) Use Taylor's Inequality to estimate the accuracy of the approximation $f(x) \approx T_2(x)$ when x lies in the interval $\underbrace{2.95 \le x \le 3.05}_{|x-3| \le 0.05}$. (You do not need to simplify your answer.)

(a) $f(3) = 8^{5/3} = 32$

$f'(x) = \frac{5}{3}(3x-1)^{2/3}(3) = 5(3x-1)^{2/3} \Rightarrow f'(3) = 5(4) = 20$

$f''(x) = \frac{10}{3}(3x-1)^{-1/3}(3) = 10(3x-1)^{-1/3} \Rightarrow f''(3) = \frac{10}{2} = 5$

$T_2(x) = 32 + 20(x-3) + \frac{5}{2!}(x-3)^2$

$= \boxed{32 + 20(x-3) + \frac{5}{2}(x-3)^2}$

(b) $f'''(x) = -\frac{10}{3}(3x-1)^{-4/3}(3) = \frac{-10}{(3x-1)^{4/3}}$

On $[2.95, 3.05]$,

$|f'''(x)| = \frac{10}{|3x-1|^{4/3}} \le \frac{10}{[3(2.95)-1]^{4/3}}$ or $\frac{10}{(7.85)^{4/3}}$

$|R_2(x)| \le \frac{\frac{10}{[3(2.95)-1]^{4/3}}}{3!}|x-3|^3$

$\le \boxed{\frac{10(0.05)^3}{3![3(2.95)-1]^{4/3}}}$ or $\frac{5(0.05)^3}{3(7.85)^{4/3}}$

5) (20 points) (a) Find the center, foci, and vertices of the ellipse and sketch the graph.

$$4x^2 + y^2 + 8x - 4y = 8$$

$4(x^2+2x)+y^2-4y=8$

$4(x^2+2x+1)+y^2-4y+4=8+4+4$

$4(x+1)^2+(y-2)^2=16$

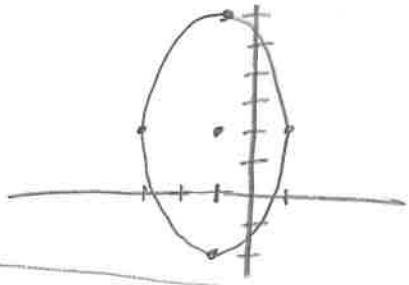
$\frac{(x+1)^2}{4}+\frac{(y-2)^2}{16}=1$

$\boxed{\text{center: } (-1,2)}$

$a=4,\ b=2$

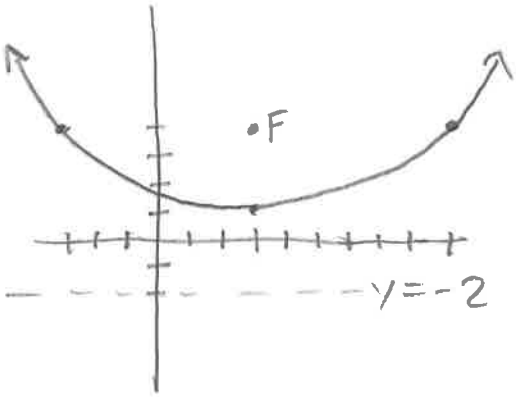
$c^2=a^2-b^2=16-4=12$

$c=2\sqrt{3}$ or $\sqrt{12}$



$\boxed{\begin{array}{l}\text{vertices: } (-1,6),(-1,-2)\\ \text{foci: } (-1,2+2\sqrt{3}),(-1,2-2\sqrt{3}) \text{ or } (-1,2\pm\sqrt{12})\end{array}}$

(b) Find an equation of the parabola with focus $(3,4)$ and directrix $y=-2$. Sketch its graph.



vertex: $(3,1)$

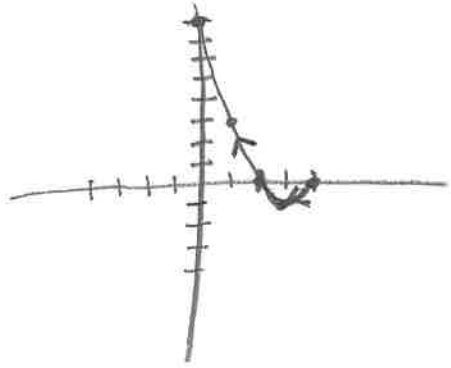
$p=3$, opens up

$\boxed{(x-3)^2=12(y-1)}$

6) **(20 points) (a) Sketch the parametric curve and eliminate the parameter to find the Cartesian equation of the curve. (Simplify your answer.)**

$$x = 4 - t,\ \ y = t^2 - 2t,\ \ 0 \le t \le 4$$

t	(x,y)
0	(4,0)
1	(3,-1)
2	(2,0)
3	(1,3)
4	(0,8)



$t = 4 - x$

$y = (4-x)^2 - 2(4-x) = 16 - 8x + x^2 - 8 + 2x$

$\boxed{y = x^2 - 6x + 8}$, $0 \le x \le 4$

(b) Find the arc length of the curve.

$$x = 4 - 3t^2,\ \ y = 2t^3 + 1,\ \ 0 \le t \le 4$$

$$L = \int_0^4 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt = \int_0^4 \sqrt{(-6t)^2 + (6t^2)^2}\, dt$$

$$= \int_0^4 \sqrt{36t^2 + 36t^4}\, dt$$

$$= \int_0^4 6t\sqrt{1+t^2}\, dt$$

$u = 1 + t^2$

$du = 2t\,dt$

$$= 3\int_1^{17} \sqrt{u}\, du$$

$$= 3 \cdot \frac{2}{3} u^{3/2} \Big|_1^{17}$$

$$= \boxed{2(17^{3/2} - 1)}$$